

CS348 Computer Networks MidSem, Sat 12 Sep 2026, 8-10am, Max. Marks: 30

Name: SAMPLE ANSWERS Roll number: br12345 House: _____

General instructions

- Write only in the given space. Answer briefly but crisply. **Strong recommendation: use DARK PENCIL**
- You are allowed to refer to your own hand-written notes only. You do NOT need a calculator.
- All answers have to be (briefly) explained. State any necessary assumptions. No clarifications during exam.
- The final answer of each question has to be entered in the given box.

Q1: [2 x 5 = 10 marks] Short answer questions

1. I once bought a wired earphone to use while watching videos. I found that if I played at twice the video speed, the video was fine but the audio was choppy/unclear only when I used the earphone (not when using the laptop speaker). What might be the reason behind this? Explain using concepts learnt in this course.

Reason: The freq. range of the audio signal doubles at 2x speed, and is likely exceeding the bandwidth of the earphone

2. Ms. H. D. Elsi is experimenting with a new flag 01010101 to use as the start/end sentinel in framing. In order to prevent data bits from being confused with the flag, she decided to stuff a one after receiving the data sequence 010101. Will this work? Justify.

Answer (Y/N):

N

Reason: If data ends in 01, then receiver sees 0101010101 and concludes frame end 2 bits earlier

3. Suppose that a communication link has 4 MHz bandwidth. What is the achievable capacity of this channel under any encoding, if the SNR is 36dB? (Make suitable engineering approximations; 3dB = factor of 2)

Answer (closest integer, in Mbps):

48

Calculation: $SNR = 36dB \approx 10^3 \times 4 = 4000$. Capacity $= 4M \times \log_2(4000+1)$
 $\approx 4M \times \log_2(2^{12}) = 48 Mbps$

4. Explain the following using concepts learnt in this course. (a) use of a flyover to reduce congestion at a road intersection, (b) the intersection at IIT main gate switches to flashing yellow lights during late night

(a): additional freq. channel, FDMA $\rightarrow$ no waiting delay, more capacity

(b): Contention-based access (CSMA) more efficient at low load

5. Among the functionalities of (i) medium access control (MAC), (ii) retransmission, (iii) forward-error-correction, (iv) error-detection, which are provided by:

(a) WiFi: MAC, retransmission, error-detection

(b) Fully Switched Ethernet: error-detection

Q2: IP fragmentation and pipelining [3+1=4 marks]: In the network below, N1 sends a packet of size P bytes (IP payload) to N2 via IP routers R1 and R2. Assume that TCP and IP headers are both 20 bytes each. Ignore propagation and processing delays, and assume that there is no other traffic in the network. Consider two scenarios: S1 where the packet is sent as a single IP fragment, and S2 where the packet is split into 3 fragments of equal size.



1. For what values of P will scenario S1 have a larger delay for the packet, compared to scenario S2 ?

Answer:

$$P > 30$$

Reason: Delay in S1: $\left(\frac{P+20}{R}\right) \times 3$ where R = link capacity
 Delay in S2: $\left(\frac{\frac{P}{3} + 20}{R}\right) \times 5$. So $(P+20) \times 3 > \left(\frac{P}{3} + 20\right) \times 5$
 $\frac{4P}{3} > 40$ or $P > 30$

2. What is the minimum value of P for which scenario S2 is possible?

Answer:

$$24$$

Calculation: P has to be a multiple of 3 to be split into 3 parts.
 $\frac{P}{3}$ has to be a multiple of 8 to be IP frag size. So 24.

Q3: Forwarding [1x4 = 4 marks]: The forwarding table at a router in the global Internet is as shown below. The prefixes/addresses are represented in hexadecimal for ease of conversion to binary.

Network Prefix	Next-hop
C4.5E.2.0/23	A
C4.5E.4.0/22	B
C4.5E.C0.0/19	C
C4.5E.40.0/18	D
C4.4C.0.0/14	E
C0.0.0.0/2	F
80.0.0.0/1	G

1. Which next hop will a packet with destination address C4.4B.31.2E be routed to?

Answer:

$$F$$

Calculation: C4.4B does not match first 14 bits of C4.4C

2. Which next hop will a packet with destination address C4.5E.05.09 be routed to?

Answer:

$$B$$

Calculation: C4.5E.05 matches first 22 bits of C4.5E.04

3. Which next hop with a packet with destination address C4.4D.31.2E be routed to?

Answer:

E

Calculation: C4.4D matches first 14 bits of C4.4E

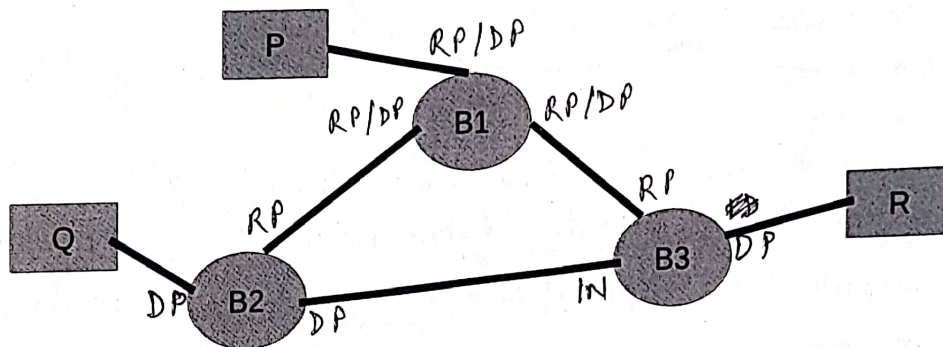
4. Which next hop with a packet with destination address C0.A8.1.2 be routed to?

Answer:

DROP

Calculation: 192.168.0.0 is a private address - dropped on global Internet

Q4: Ethernet [2+3+2=7 marks]: In the switched Ethernet below, B1, B2, B3 are bridges; P, Q, R are end-hosts.



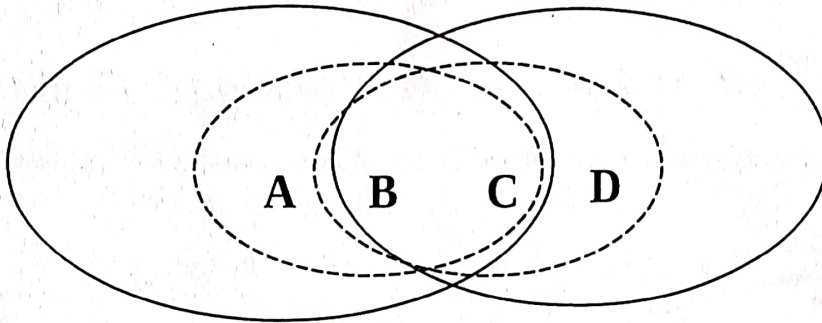
- Suppose the spanning tree protocol has stabilized. In the above diagram itself, mark each port of each bridge as RP (root port), DP (designated port), or IN (inactive).
- Suppose that IN (inactive port) means that no packet will be sent or received on that port. Construct a short sequence of events E1, E2, E3... (as many as needed) to show what *undesirable* thing can happen with such a definition of IN.

Answer: E1: B1 fails, E2: B2 announces itself to be root
Announcement never heard by B3 if IN does not receive
any packet at all. Hosts Q & R cannot talk even
though path exists - undesirable.

- Given the above, what exactly *does* marking a port as IN (inactive) mean? What exactly can or cannot be sent/received on such a port?

Answer: Data packets are not sent or received on IN
STP packets are sent/received
Spanning Tree Protocol

Q5: Wireless and MAC [0.5+0.5+1+1+2=5 marks]: Consider the wireless network below. The transmit ranges of each node is as shown. A can be heard by B and C but not by D. B can be heard by A and C but not by D. C can be heard by B and D but not by A. D can be heard by B and C but not by A. Suppose A is sending data to B and C is sending data to D. Both are sending packets as fast as is allowed by the MAC. There is no other traffic. Ignore header overheads and propagation and processing delays.



1. Mention one hidden node scenario and one exposed node scenario in the above network.

Hidden node scenario: D & A hidden from each other w.r.t B or C

Exposed node scenario: B & C exposed to each other when tx to A & D resply.

2. Suppose no MAC is employed i.e. there is nothing to detect or avoid collisions. There are no ACKs either. What is the received throughput at B as a percentage of A's sending rate?

Answer (percentage, nearest integer): 0

Reason: All frames lost due to interference from C

3. Now suppose that each node uses CSMA/CA with a fixed contention window. There is no RTS/CTS or link layer ACK/retransmission. What is the received throughput at B as a percentage of A's sending rate?

Answer (percentage, nearest integer): 50 End of every frame at C, 50% chance

Reason: A wins (no collision), 50% chance C wins (collision)

4. In the same situation as above, what is the received throughput at D as a percentage of C's sending rate?

Answer (percentage, nearest integer): 100

Reason: No interference from A at D, so all frames received

5. Now consider that each node uses CSMA/CA with a fixed contention window. There is no RTS/CTS, but there is WiFi-like immediate ACK and up to 1 retry (so 2 tries including original transmission). What is the received throughput at D as a percentage of the link speed?

Answer (percentage, nearest integer): 25

Reason: 50% of link capacity C transmits, but all ACKs from D lost due to collision from A. so all frames 2 tries.